Computer Organization
Homework 1
(Due on 2018/10/4)

1. Assume a 15cm diameter wafer has a cost of 12, contains 84 dies, and has 0.020 defects/cm^2. Assume a 20 cm diameter wafer has a cost of 15, contains 100 dies, and has 0.035 defects/cm^2.
   (a) Find the yield for both wafers (10%)
   (b) For the 20cm diameter wafer, if the number of dies per wafer increased by 10% and the defect per area unit increases by 10%. Find the die area and yield. (10%)
2. The A processor has a clock rate of 3.2GHz and voltage of 1.2V. Assume that, on average, it consumed 10W of static power and 80W of dynamic power.
   The B processor has a clock rate of 4 GHz and voltage of 0.8V. Assume that, on average, it consumed 20W of static power and 50W of dynamic power.
   (a) For each processor, find the average capacitive loads. (10%)
   (b) Find the percentage of the total dissipated power comprised by static power and the ratio of static power to dynamic power for each processor. (10%)
   (c) If the total dissipated power is to be reduced by 10%, how much should the voltage be reduced for two processors to maintain the same leakage current? (Assume static power=leakage current * voltage) (10%)
3. Consider two different implementations of the same instruction set architecture. The instructions can be divided into five classes according to their CPI (class A, B, C, D, and E), P1 with a clock rate of 2GHz and CPIs of 1,2,3,4,5, and P2 with a clock rate of 2.5GHz and CPI of 2,3,3,4,4. Given a program with a dynamic instruction count of 1.0e8 instructions divided into classes as follows: 20% class A, 25% class B, 30% class C, 10% class D, and 15% class E.
   (a) Which implementation is faster? Please give the reason. (10%)
   (b) What is weighted average CPI (global CPI) for each implementation? (10%)
   (c) Find the clock cycles required in both cases. (10%)
4. Assume for arithmetic, load/store and branch instructions, a processor has CPIs of 1, 12, and 5, respectively. Also assume that on a single processor a program requires the execution of 2.56×10^9 arithmetic instructions, 1.28×10^9 load/store instructions, and 256 million branch instructions.
   (a) BY how much must we improve the CPI of load/store instructions if we want the program to run two times faster? (10%)
   (b) By how much is the execution time of the program improved if the CPI of arithmetic and branch instructions is reduced by 50% and the CPI of load/store is reduced by 20%? (10%)